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EDITOR'S NOTE

The AHA/ACC Guideline for Pulmonary Embolism

A New Era in Diagnosis, Risk Assessment, and Management

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In this issue of *JACC*, the 2026 AHA/ACC/ACCP/ACEP/CHEST/SCAI/SHM/SIR/SVM/SVN Guideline for the Evaluation and Management of Acute Pulmonary Embolism in Adults: (hereafter referred to as the 2026 AHA/ACC Acute PE guideline), the first AHA/ACC formal guideline document for venous thromboembolism, is published.[1] Considering the high burden of PE—with tens of thousands of annual hospitalizations and deaths in the United States but also globally[2-4]—as well as advances in diagnostics, risk assessment, drug and device therapy, and follow-up of patients with PE, this is a very welcome addition to the professional society recommendations.[5] Herein, we summarize a selection of the key recommendations through a hypothetical patient's journey from risk factors to diagnosis, short- and long-term management, and follow-up (**Figure 1**), and we reflect on areas of future clarification or update.

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Diagnostic recommendations in the 2026 AHA/ACC Acute PE guideline emphasize history and physical examination combined with age-adjusted D-dimer testing and chest imaging (most often computed tomography pulmonary angiography) in patients with high (>50%) probability of PE.[1] The guideline shares a clear path for pregnant individuals suspected to have PE, with distinct considerations for pregnancy-adapted clinical criteria, D-dimer testing, and chest imaging.[6]

The single most noticeable development in the 2026 AHA/ACC Acute PE guideline compared with prior professional society recommendations is the AHA/ACC Acute PE Clinical Categories. While building on prior classifications such as that of the European Society of Cardiology (low, intermediate, and high risk), the Clinical Categories incorporate other considerations to rectify the full spectrum of illness severity from asymptomatic presentations (AHA/ACC Acute PE Category A) to fulminant cardiorespiratory collapse (AHA/ACC Acute PE Category E). Similar to prior schemes, the AHA/ACC Acute PE Clinical Categories embrace vital signs, standardized clinical risk assessment (via either the pulmonary embolism severity index, simplified pulmonary embolism severity index, or Hestia criteria), cardiac biomarker testing, and assessment of right ventricular function. However, the 2026 AHA/ACC Acute PE guideline incorporates data elements such as lactate or invasively assessed cardiac index to characterize incipient (normotensive) shock. In addition, for the first time, there is clear recognition of respiratory modifiers (R) in the Clinical Categories.

The recommendations for drug, device, and ancillary therapy are based on the AHA/ACC Acute PE Clinical Category at the time of assessment. For Categories A and B (low risk of adverse events), up-

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front utilization of a direct oral anticoagulant with outpatient management or early discharge is considered. For patients requiring admission for parenteral therapy, low-molecular weight heparin is favored over unfractionated heparin. For Category E (highest risk of death), the 2026 AHA/ACC Acute PE guideline favors advanced therapies including systemic or catheter-directed fibrinolysis,[7] mechanical thrombectomy, and surgical embolectomy. For those with Categories C and D, the guideline shares suggestions for when these options might be considered.

> *"The single most noticeable development in the 2026 AHA/ACC Acute PE guideline compared with prior professional society recommendations is the AHA/ACC Acute PE Clinical Categories"*

Another key element emphasized in the 2026 AHA/ACC Acute PE guideline is that care of patients with PE does not stop after the acute event. Follow-up at 3 to 6 months to evaluate for post-PE syndrome and to weigh the risks, benefits, and patient values for continued anticoagulation, in particular for those with persistent risk factors including cancer, are critical. Additionally, there is an emphasis on early ambulation and detection of the spectrum of post-PE syndrome, including chronic thromboembolic pulmonary hypertension and its management at centers of excellence.

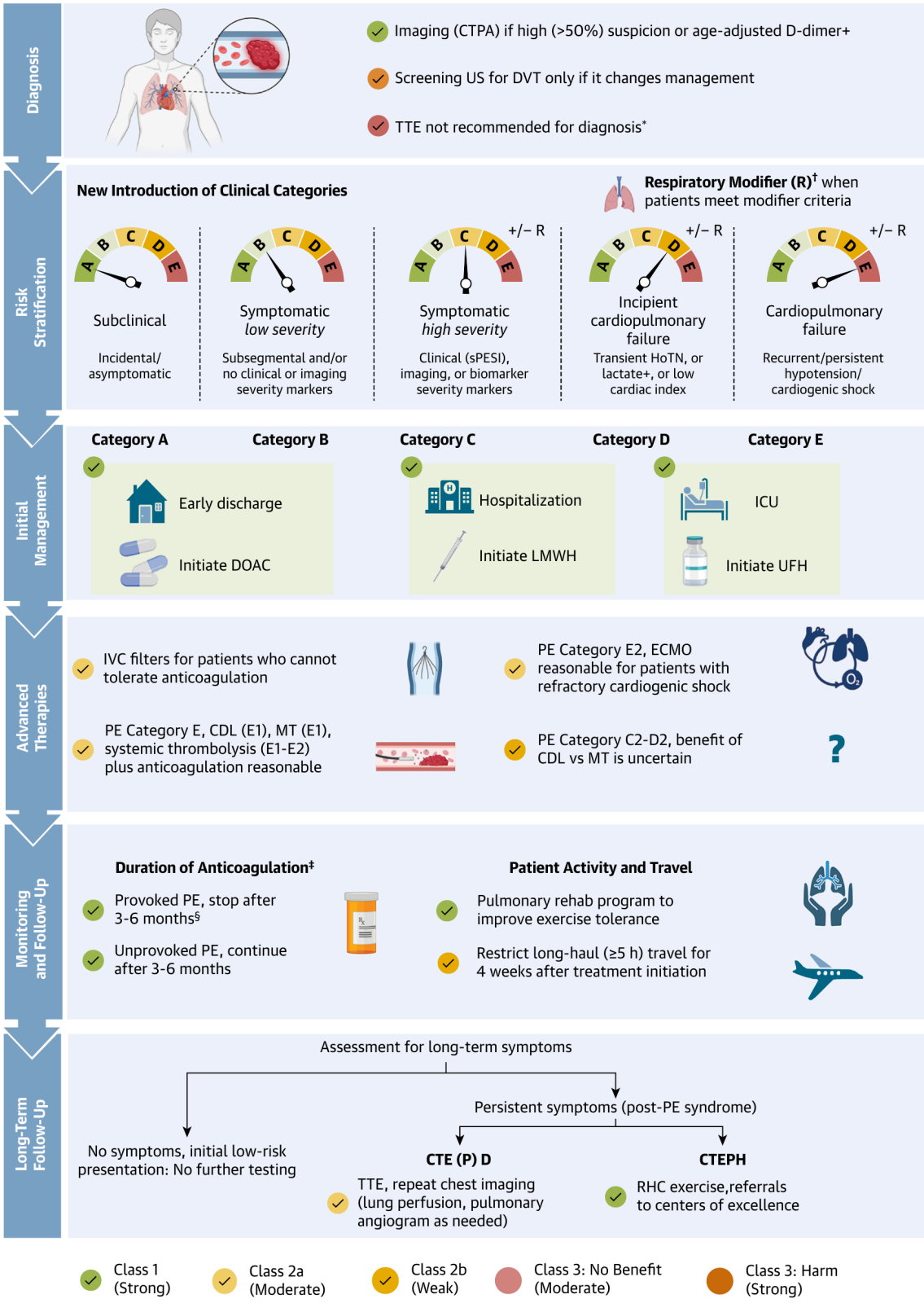
The 2026 AHA/ACC Acute PE guideline marks a major advancement for PE care. With increasing involvement of cardiovascular medicine specialists in managing PE, a new cardiovascular guideline in North America is timely, similar to what has been ongoing in Europe. Further, the document presents evidence-driven recommendations for diagnosis, as well as pharmacological and non-pharmacological management considerations. Also, the AHA/ACC Clinical Categories provide a comprehensive categorization with the potential to optimize care based on overlooked factors such as respiratory status or early markers of hemodynamic instability.

One area that clinicians may find particularly challenging is translating guideline recommendations into practice—specifically deciding whether endovascular therapy should be used in a hemodynamically stable, non-low-risk patient, the timing of such intervention, and selecting the intervention, such as catheter-directed fibrinolysis vs mechanical thrombectomy. These uncertainties stem from limitations in the existing evidence. Efforts are needed to develop risk-stratification tools with higher positive predictive value than the current 5% to 15% for identifying patients at risk of death or decompensation on anticoagulant monotherapy. Results from ongoing clinical trials are expected to inform the timing and selection of endovascular strategies, enabling future recommendations to be refined and made more precise.

There are also opportunities for future improvements, either by complementary concepts or through updates as the knowledge base for PE expands with high-quality studies in epidemiology and comparative effectiveness/safety of therapies. The AHA/ACC Acute PE Clinical Categories, although intuitive and encompassing an array of clinical factors and biomarkers, benefit from validation against existing tools, particularly to understand the positive predictive value for consideration of patients for advanced therapies. Similarly, although the 2026 AHA/ACC Acute PE guideline does not specify troponin assay type, recent evidence suggests that high-sensitivity troponin—now the dominant existing assays in U.S. practice—may not substantively improve risk stratification[8] and may lead to unnecessary advanced therapy. When and how thrombophilia testing should be considered is also an area of future updates. Although the guideline recommendations are reasonable based on current evidence, it is important to note that restricted testing for thrombophilia is largely driven by absence of robust evidence, rather than evidence of futility of testing, for example, for antiphospholipid antibodies.[9] For acute care, whereas team-based approaches are intuitive, much of the existing evidence is of limited methodological quality. Cluster-randomized trials can demonstrate whether these teams and their impact on procedural therapy-use improve patient outcomes.[10] Additionally, high-quality investigations are needed to guide us on optimal post-PE exercise and dietary recommendations and for management of post-PE syndrome. Finally, future updates can share guidance on key subgroups, particularly by sex, among elderly populations, and patients with cancer.

The 2026 AHA/ACC Acute PE guideline marks a pivotal advance in the management of PE in the United States. With enduring risk factors such as cancer and obesity on the rise and the expanding role

FIGURE 1 **Graphical Summary of Recommendations in the 2026 ACC/AHA Guideline for PE From Diagnosis to Follow-Up**



Continued on the next page

of cardiovascular medicine specialists, this evidence-based guideline is timely in shaping multidisciplinary care. Although the title is framed around acute PE, the text rightly underscores that the impact of PE extends beyond the initial event. Longitudinal management has become essential to optimize patient outcomes and reduce the long-term burden of this complex disease.

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FIGURE 1 Continued

*Although transthoracic echocardiography (TTE) is not recommended for diagnosis, it can inform pulmonary embolism (PE) risk stratification. †The respiratory modifier (R) is added when a patient with acute PE has notable respiratory compromise such as hypoxemia, tachypnea, or a need for supplemental oxygen. It highlights prominent respiratory abnormalities within any PE category. ‡Although the guideline shares a summary of several relevant studies, it does not share direct guidance about the duration of anticoagulation in patients with cancer-associated PE and does not specify a preferred agent/dose beyond suggesting either low molecular weight heparin (LMWH) or direct oral anticoagulants (DOACs). §More recent evidence reported by Piazza et al indicates that even in those with provoked PE, durable low-intensity anticoagulation may be reasonable if enduring risk factors are present. (Piazza G, Bikdeli B, Pandey AK, et al, HI-PRO Trial Investigators. Apixaban for extended treatment of provoked venous thromboembolism. *N Engl J Med*. 025;393(12):1166-1176.) CDL = catheter-directed thrombolysis; CTE(P)D = chronic thromboembolic pulmonary disease; CTEPH = chronic thromboembolic pulmonary hypertension; CTPA = computed tomography pulmonary angiography; DVT = deep vein thrombosis; ECMO = extracorporeal membrane oxygenation; HoTN = hypotension; ICU = intensive care unit; IVC = inferior vena cava; MT = mechanical thrombectomy; RHC = right heart catheterization; sPESI = simplified pulmonary embolism severity index; UFH = unfractionated heparin; US = ultrasound.

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